SECSR2043 OPERATING SYSTEMS
[20 Marks]

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Section	:	02	

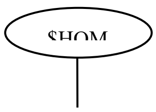
Instruction: Please answer all of the following questions. Whenever the symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname*.sh (Example: `ahmad.sh`).

```
echo “Hello world” > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo “Honda Accord” > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.* .; cp ../fruits/*.* .;
cp ../*.jar .
```

   a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **$HOME** directory.

   [4 marks]



Print screen the script that you type;

Then draw the tree

Script Typed:

```
#!/bin/bash

echo "Hello world" > helloworld.jar

mkdir cars
mkdir dates
mkdir fruits drinks

cd cars
echo "Honda Accord" > accord.c
cp accord.c civic.c
echo proton > proton.c
cd ../dates
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits
echo apple > apple.txt
cat apple.txt > orange.txt
cd ../drinks
cp ../cars/*.* .
cp ../fruits/*.* .
cp ../*.jar .
```

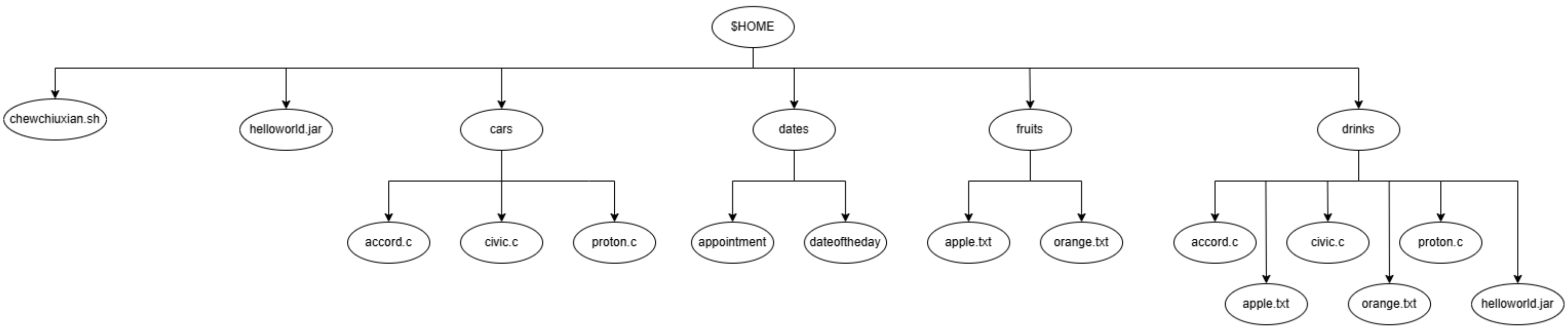
Code Typed:

```
nano chewchiuxian.sh
chmod +x chewchiuxian.sh
./chewchiuxian.sh
```

Tree Structure:

```
.
|-- cars
|   |-- accord.c
|   |-- civic.c
|   `-- proton.c
|-- chewchiuxian.sh
|-- dates
|   |-- appointment
|   `-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   `-- proton.c
|-- fruits
|   |-- apple.txt
|   `-- orange.txt
`-- helloworld.jar
```

Tree Map:



b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter "c" as the input to the bash script. [4 marks]

Script Typed:

```
#!/bin/bash

read -p "Enter the file extension: " ext
echo "Files with .$ext extension:"
find . -type f -name "*.$ext"
count=$(find . -type f -name "*.$ext" | wc -l)
echo "Total .$ext files: $count"
```

Script Run:

```
chew@secr2043:~$ nano chewchiuxian2.sh
chew@secr2043:~$ chmod +x chewchiuxian2.sh
chew@secr2043:~$ ./chewchiuxian2.sh
Enter the file extension: c
Files with .c extension:
./os_lab/echo.c
./os_lab/sum.c
./os_lab/hello.c
./workspace/hello_world.c
./workspace/arguement.c
./drinks/civic.c
./drinks/accord.c
./drinks/proton.c
./fork_wait_example.c
./subprocess1.c
./subprocess2.c
./cars/civic.c
./cars/accord.c
./cars/proton.c
./mainprocess.c
Total .c files: 15
```

2. The following Figure 1 illustrates a tree structure of some directories and files.

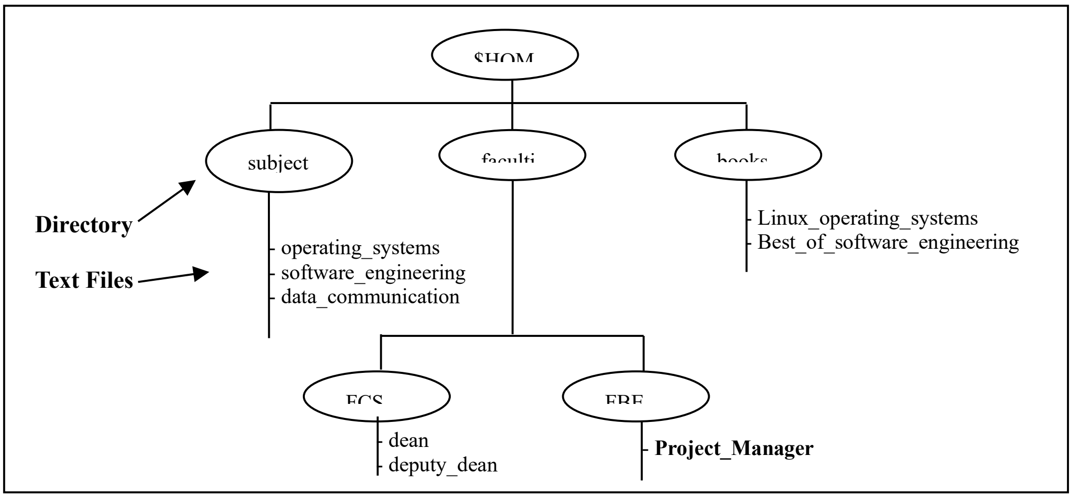


Figure 1

a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character**.** For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Script Typed:

```
#!/bin/bash

mkdir -p $HOME/subjects $HOME/faculties/FCS $HOME/faculties/FBE $HOME/books

# Subjects directory
echo "operating systems" > $HOME/subjects/operating_systems
echo "software engineering" > $HOME/subjects/software_engineering
echo "data communication" > $HOME/subjects/data_communication

# Faculties/FCS directory
echo "dean" > $HOME/faculties/FCS/dean
echo "deputy dean" > $HOME/faculties/FCS/deputy_dean

# Faculties/FBE directory
echo "Project Manager" > $HOME/faculties/FBE/Project_Manager

# Books directory
echo "Linux operating systems" > $HOME/books/Linux_operating_systems
echo "Best of software engineering" > $HOME/books/Best_of_software_engineering
```

Script Run:

```
|-- subjects
|   |-- data_communication
|   |-- operating_systems
|   `-- software_engineering
|-- books
|   |-- Best_of_software_engineering
|   |-- Linux_operating_systems
|-- faculties
|   |-- FBE
|   |   `-- Project_Manager
|   `-- FCS
|       |-- dean
|       `-- deputy_dean
```

b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called `book`. [2 marks]

Directory/File	Access Control
books	drwxrwxr-x
subjects	drwxr-xr-x
Best_of_software_engineering	-rw-r--r--
FCS	drwxr-xr-x
project_manager	-rw-r--r--

c) Write another bash script (called `myname2c.sh`) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
subjects	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓

Script Typed:

```
# !/bin/bash

chmod 744 $HOME/subjects
chmod 520 $HOME/books/Best_of_software_engineering
chmod 607 $HOME/faculties/FCS
chmod 031 $HOME/faculties/FBE/Project_Manager
```

Script Run:

```
drwxr--r-- 2 chew chew  4096 Jun 17 04:03 subjects
-r-x-w---- 1 chew chew   29 Jun 17 04:03 Best_of_software_engineering
drw----rwx 2 chew chew 4096 Jun 14 08:53 FCS
-----wx--x 1 chew chew 16 Jun 17 04:03 Project_Manager
```

d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subjects	drwxr-r--
Best_of_software_engineering	-r-x-w----
FCS	drw----w-x
project_manager	-----wx--x

End of Lab 3

**** All the Best for Final Exam ****